

ENGE702 Engineering Mathematics III

Tutorial 4

1. Find the general solution to the following differential equations:

$$y = C_1 e^{-4x} + C_2 e^{3x} - \frac{4}{65} \sin(2x) - \frac{1}{130} \cos(2x) \quad y = C_1 e^{-4x} + C_2 e^{2x} + \frac{5}{6} x e^{2x}$$

(a) $y'' + y' - 12y = \sin(2x)$ (c) $y'' + 2y' - 8y = 5e^{2x}$

$$y = C_1 e^{-4x} + C_2 x e^{-4x} + \frac{3}{16} x^2 - \frac{3x}{16} + \frac{9}{128}$$

(b) $y'' + 8y' + 16y = 3x^2$ (d) $y'' + 4y' + 6y = 7$

$$y = e^{-2x} (A \sin(\sqrt{2}x) + B \cos(\sqrt{2}x)) + \frac{7}{6}$$

2. Solve the following initial value problems:

(a) $y'' + 4y' + 3y = 3x$, $y(0) = 2$, $y'(0) = -1$. $y = -\frac{2}{3} e^{-3x} + 4e^{-x} + x - \frac{4}{3}$

(b) $y'' - 4y' + 4y = \cos(x)$, $y(0) = 3$, $y'(0) = 0$. $y = \frac{72}{25} e^{2x} - \frac{28}{5} x e^{2x} - \frac{4}{25} \sin(x) + \frac{3}{25} \cos(x)$

(c) $y'' + 2y' + 6y = e^{3x}$, $y(0) = 3$, $y'(0) = -2$. $y = \frac{17\sqrt{5}}{105} e^{-x} \sin(\sqrt{5}x) + \frac{62}{21} e^{-x} \cos(\sqrt{5}x) + \frac{1}{21} e^{3x}$

3. Find a function to describe the displacement in a forced spring mass damper system with the following parameters (this will be a general solution so there will be arbitrary constants):

(a) $m = 1$, $c = 4$, $k = 3$, $r(t) = 3 \cos(3t)$. $y = C_1 e^{-3t} + C_2 e^{-t} + \frac{1}{5} \sin(3t) - \frac{1}{10} \cos(3t)$

(b) $m = 1$, $c = 0$, $k = 4$, $r(t) = 5 \cos(2t)$. $y = C_1 \cos(2t) + C_2 \sin(2t) + \frac{5}{4} t \sin(2t)$

4. Suppose we have a forced spring mass damper system with the parameters $m = 1$, $c = 0$, $k = 9$, $r(t) = 5 \sin(\omega t)$ for some $\omega \neq 3$.

- (a) Find an initial condition (ie a value for $y(0)$ and $y'(0)$) that will give a particular solution of:

$$y = \frac{F_0}{m(\omega_0^2 - \omega^2)} \cos(\omega t) - \frac{F_0}{m(\omega_0^2 - \omega^2)} \cos(\omega_0 t) \quad y(0) = 0, y'(0) = 0 \text{ will do.}$$

$(\frac{\omega_0 - \omega}{2})T = \pi$ (2 beats per 2π)

- (b) What value of ω would give a 'beat period' of $T_b = 5$. The beat period is the time between the highest amplitudes in a system experiencing beats. Note that $\omega = \frac{2\pi}{T}$.
(note a full period of sin is 2 beats) $\omega = 3 - \frac{4\pi}{10} \approx 1.74$

5. For each of the following systems, find a value for ω that will give the largest steady state amplitude of oscillations, and find that amplitude.

(a) $m = 3$, $c = 1$, $k = 5$, $r(t) = 3 \sin(\omega t)$. $\omega_{max} = 1.27$ $C^*(\omega_{max}) = 2.34$

(b) $m = 1$, $c = 2$, $k = 3$, $r(t) = \cos(\omega t)$. $\omega_{max} = 1$ $C^*(\omega_{max}) = 0.35$

6. Calculate the steady state amplitude in the following systems:

(a) $m = 2$, $c = 3$, $k = 3$, $r(t) = 3 \sin(3t)$. $C^* = 0.175$

(b) $m = 5$, $c = 2$, $k = 4$, $r(t) = 2 \sin(\pi t)$. $C^* = 0.044$

4b. $T_b = 5$, $T = 2T_b$ for sine/cosine function.
 $\frac{2\pi}{2T_b}$ is the frequency $\frac{\omega_0 - \omega}{2}$

Computer Part

Even though tutorials are not run in computer labs, during weeks 1 - 6 of the semester, you should work through the section 'Solving Ordinary Differential Equations With MATLAB' (under 'Computational Mathematics') in the self paced courses on the mathworks website, as 'homework'.

If you have never used matlab before, it would be a good idea to work through the section 'MATLAB Onramp' first. Details of how to access these are as follows:

AUT has a licence with the matlab creators that allows you to access an array of online training material, and also download the latest version of matlab to your personal computer (if you wish - you don't need to do this for this course if you don't want to, you can access matlab on the computers at AUT).

To access these resources you will need to create an account with mathworks (the company that produces matlab).

You can do this by visiting [mathworks.com](https://www.mathworks.com) (or google 'matlab'). You can create an account by clicking the 'login' icon on the top right. **It is important you use your AUT email address to create the account, or you will not be able to access the material.**

From your account page, click 'Self-Paced Courses'.